Euclidean Norm Maximization over Zonotopes is W[1]-hard

1 Introduction

A zonotope is a Minkowski sum of line segments, $Z = \sum_{\ell=1}^{m}[0, g_\ell] = \left\{\sum_{\ell=1}^{m} \theta_\ell g_\ell : \theta_\ell \in [0,1]\right\} \subseteq \mathbb{R}^d$, specified by its generators $g_1, \dots, g_m \in \mathbb{R}^d$. Let $p \geq 1$, L_p-Norm-Maximization asks for $\max_{z\in Z} \|z\|_p$. The problem is NP-hard in general but solvable in polynomial time for every fixed dimension d, by brute force over the $O(m^{d-1})$ vertices. A COLT 2025 open question [4] asked whether it is *fixed-parameter tractable* (FPT) in d, i.e. solvable in time $f(d) \cdot m^{O(1)}$ for some computable f. In this paper, we resolve the Euclidean case:

Theorem 1. *Let $Z = \sum_{\ell=1}^{m}[0, g_\ell] \subseteq \mathbb{R}^d$ be a zonotope and T be a rational threshold, deciding whether $\max_{z\in Z} \|z\|_2 \geq T$ is* W[1]*-hard parameterized by the dimension d.*

Related work. The work of Froese, Grillo, Hertrich and Stargalla [5] proved W[1]-hardness (parameterized by d) of *zonotope non-containment*, which via the equivalence of [7] gives W[1]-hardness of maximizing a *polyhedral norm* over a zonotope. The smooth case $p \in (1, \infty)$ was explicitly left open (see the conclusion section in [5]), and the gap is structural: non-containment hardness relies on a polytopal unit ball, whereas for $p \in (1, \infty)$ the unit ball is smooth and strictly convex, so the polyhedral gadgets do not transfer.

For polytopes given in halfspace H-presentation, Knauer, König and Werner [6] have proved L_p-maximization is W[1]-hard in d for every $p \in \mathbb{N} \setminus \{1\}$ (and FPT for $p = 1$), building on [2]. This does not carry over to our setting: a zonotope with m generators can have $m^{\Theta(d-1)}$ facets, so converting the generator-given form to an H-presentation is not an FPT reduction, and the two representations are parameterized-incomparable. Indeed, on zonotopes L_∞-maximization is polynomial-time solvable and L_1-maximization is FPT in d [1, 5]; our result shows this tractability does not extend to the Euclidean norm.

1.1 Background

Parameterized complexity A parameterized problem is *fixed-parameter tractable* (FPT) if it is decidable in time $f(k) \cdot n^{O(1)}$ for input size n and parameter k. A *parameterized reduction* from (L, k) to (L', k') is a map computable in time $f(k) \cdot n^{O(1)}$ that preserves membership and satisfies $k' \leq g(k)$ for computable f, g; such a reduction from a W[1]-hard problem shows W[1]-hardness, and no W[1]-hard problem is FPT unless FPT = W[1]. Our source problem is:

Theorem 2 ([3])**.** *Multicolored Clique is* W[1]*-hard parameterized by k: given a graph G whose vertex set is partitioned into color classes $V_1, \dots, V_k$, decide whether G has a clique containing exactly one vertex from each class.*

Throughout, N denotes the maximum color-class size, and we identify each V_i with a subset of $[N]$.

Support functions and Euclidean duality The *support function* of a convex body $K \subseteq \mathbb{R}^d$ is $h_K(u) = \max_{z\in K}\langle u, z\rangle$; it is additive under Minkowski sums. For a segment, $h_{[0,g]}(u) = [\langle g, u\rangle]_+$ with $[t]_+ = \max\{t, 0\}$, so a zonotope $Z = \sum_\ell [0, g_\ell]$ has $h_Z(u) = \sum_{\ell=1}^m [\langle g_\ell, u\rangle]_+$. Since $\|z\|_2 = \max_{\|u\|_2=1}\langle u, z\rangle$, swapping the two maxima gives the duality

$$\max_{z\in Z}\|z\|_2 = \max_{\|u\|_2=1} h_Z(u) = \max_{\|u\|_2=1} \sum_{\ell=1}^{m} [\langle g_\ell, u\rangle]_+ . \tag{1}$$

We also use that a centrally symmetric convex polygon $P = \mathrm{conv}\{\pm v_1, \ldots, \pm v_N\} \subseteq \mathbb{R}^2$ with $2N$ vertices is a *zonogon*: its boundary edges come in opposite parallel pairs, and choosing one edge vector z_ℓ from each of the N pairs (rational when the v_r are) gives

$$P = \sum_{\ell=1}^{N} [-\tfrac{1}{2}z_\ell, \tfrac{1}{2}z_\ell] \tag{2}$$

and this is *exactly* the generator-given zonotope with the $2N$ generators $\pm\frac{1}{2}z_\ell$.

2 The reduction

Fix an instance $(G, V_1, \ldots, V_k)$ of Multicolored Clique. We build a zonotope in dimension $d = 2k+1$, with coordinates written $u = (b, y_1, \ldots, y_k)$, where $b \in \mathbb{R}$ is a scalar bias coordinate and $y_j \in \mathbb{R}^2$ is a two dimensional encoding of a chosen vertex of color j.

Section 2.1 presents the basic two dimensional encoding, Section 2.2 describes the vertex gadget, Section 2.3 describes the edge gadget. Section 2.4 presents the overall reduction (edge gadget + vertex gadget) and proves the correctness of the reduction.

2.1 Two dimensional encoding

For $r = 1, \ldots, N$ set $t_r = \frac{r}{N+1} \in (0, 1)$ and

$$s_r = \left(\tfrac{1-t_r^2}{1+t_r^2},\ \tfrac{2t_r}{1+t_r^2}\right) \in \mathbb{Q}^2, \qquad s_{r+N} = -s_r \quad (r = 1, \ldots, N). \tag{3}$$

Lemma 3. *The $2N$ points $s_1, \ldots, s_{2N}$ are rational unit vectors, and for all distinct $r, s \in [2N]$,*

$$1 - \langle s_r, s_s\rangle \ \geq\ \mu, \qquad \mu = \frac{1}{2(N+1)^2}.$$

Proof. Since $(1-t^2)^2 + (2t)^2 = (1+t^2)^2$, each s_r in (3) is a unit vector; rationality is clear. A direct computation gives, for $r, s \in [N]$,

$$1 - \langle s_r, s_s\rangle = \frac{2(t_r - t_s)^2}{(1+t_r^2)(1+t_s^2)}.$$

For distinct $r, s \in [N]$ we have $|t_r - t_s| \geq \frac{1}{N+1}$ and $(1+t_r^2)(1+t_s^2) < 4$ (as $t_r, t_s < 1$), so $1 - \langle s_r, s_s\rangle > \frac{2}{4(N+1)^2} = \mu$. The remaining cases follow from $s_{r+N} = -s_r$ and for opposite-sign pairs $1 - \langle s_r, -s_s\rangle = 1 + \langle s_r, s_s\rangle \geq 1 > \mu$. ☐

Let

$$P = \mathrm{conv}\{s_1, \ldots, s_{2N}\},$$

P is a centrally symmetric rational polygon, hence a 2-dimensional zonotope. Its support function is $h_P(y) = \max_{r\in[2N]}\langle s_r, y\rangle$, so

$$h_P(y) \leq \|y\|_2, \quad \text{with equality iff } y \text{ is a nonnegative multiple of some } s_r. \tag{4}$$

2.2 The base gadget

Fix the rationals

$$\beta = \frac{4k-1}{4k+1}, \qquad \rho = \frac{4}{4k+1}, \qquad \text{so that} \qquad \beta^2 + k\rho^2 = 1.$$

For any $u \in \mathbb{R}^{2k+1}$, we write $u = (b, y_1, \dots, y_k)$, where $b \in \mathbb{R}$ and $y_j \in \mathbb{R}^2$ $(j \in [k])$, define

$$B(u) = \beta \cdot [b]_+ + \rho \cdot \sum_{j=1}^{k} h_P(y_j). \tag{5}$$

B is a nonnegative combination of zonotope support functions (see Eq. (2)), hence itself the support function of a rational zonotope.

Define

$$c(t_1, \dots, t_k) = \big(\beta,\ \rho s_{t_1}, \dots, \rho s_{t_k}\big), \qquad t_j \in [2N],$$

a candidate, and let

$$\mathcal{C} = \{c(t_1, \dots, t_k) : t_j \in [2N]\}$$

be the (finite) candidate set. Every candidate is a unit vector: $\|c\|_2^2 = \beta^2 + k\rho^2 = 1$.

Lemma 4. *For every u with $\|u\|_2 = 1$ we have $B(u) \le 1$, with equality if and only if $u \in \mathcal{C}$.*

Proof. Put $x = ([b]_+, \|y_1\|_2, \dots, \|y_k\|_2)$ and $w = (\beta, \rho, \dots, \rho) \in \mathbb{R}^{k+1}$. By (4), $B(u) \le \beta\,[b]_+ + \rho \sum_j \|y_j\|_2 = \langle w, x\rangle$. Now $\|w\|_2^2 = \beta^2 + k\rho^2 = 1$ and $\|x\|_2^2 = [b]_+^2 + \sum_j \|y_j\|_2^2 \le b^2 + \sum_j \|y_j\|_2^2 = \|u\|_2^2 = 1$, so $\langle w, x\rangle \le \|w\|_2 \|x\|_2 \le 1$ by Cauchy–Schwarz. Equality throughout requires: $\|x\|_2 = 1$, i.e. $[b]_+ = |b|$, i.e. $b \ge 0$; Cauchy–Schwarz equality with $w > 0$ componentwise, i.e. $x = w$ (a positive multiple, normalized to equal norm), giving $b = \beta$ and $\|y_j\|_2 = \rho$ for all j; and equality in (4) for each j, i.e. $y_j = \rho s_{t_j}$ for some t_j. These conditions hold exactly when $u = c(t_1, \dots, t_k)$. $\square$

Lemma 5 (Base gap)**.** *For every u with $\|u\|_2 = 1$,*

$$1 - B(u) \ \ge\ \tfrac{1}{20}\,\mathrm{dist}(u, \mathcal{C})^2,$$

where $\mathrm{dist}(u, \mathcal{C}) = \min_{c \in \mathcal{C}} \|u - c\|_2$.

Proof. Case $b \ge 0$. With x, w as in Lemma 4, now $[b]_+ = b$ and $\|x\|_2 = \|u\|_2 = 1$. For each j with $y_j \ne 0$ pick $t_j \in \arg\max_r \langle s_r, y_j\rangle$ and set $m_j = \langle s_{t_j}, y_j\rangle / \|y_j\|_2 \le 1$ (for $y_j = 0$ the j-th term below vanishes and we pick t_j arbitrarily). Let $c = c(t_1, \dots, t_k)$. Writing $r_j = \|y_j\|_2$,

$$B(u) = \beta b + \rho \sum_j r_j m_j = \langle w, x\rangle - \rho \sum_j r_j (1 - m_j),$$

hence $1 - B(u) = (1 - \langle w, x\rangle) + \rho \sum_j r_j (1 - m_j)$. Since $\|w\|_2 = \|x\|_2 = 1$,

$$1 - \langle w, x\rangle = \tfrac{1}{2}\big(\|w\|_2^2 + \|x\|_2^2 - 2\langle w, x\rangle\big) = \tfrac{1}{2}\|x - w\|_2^2.$$

Expanding both sides and using $\langle s_{t_j}, y_j\rangle = r_j m_j$,

$$\tfrac{1}{2}\|x - w\|_2^2 + \rho \sum_j r_j (1 - m_j) = \tfrac{1}{2}(b - \beta)^2 + \tfrac{1}{2}\sum_j \big(r_j^2 - 2\rho r_j m_j + \rho^2\big) = \tfrac{1}{2}\|u - c\|_2^2.$$

Therefore $1 - B(u) = \frac{1}{2}\|u - c\|_2^2 \ge \frac{1}{2}\,\mathrm{dist}(u, \mathcal{C})^2 \ge \frac{1}{20}\,\mathrm{dist}(u, \mathcal{C})^2$.

Case $b < 0$. Then $[b]_+ = 0$, so $B(u) = \rho \sum_j h_P(y_j) \le \rho \sum_j \|y_j\|_2 \le \rho\sqrt{k}\sqrt{\sum_j \|y_j\|_2^2} \le \rho\sqrt{k}$. Now $\rho\sqrt{k} = \frac{4\sqrt{k}}{4k+1} \le \frac{4}{5}$ because $4k + 1 - 5\sqrt{k} = (4\sqrt{k} - 1)(\sqrt{k} - 1) \ge 0$ for $k \ge 1$. Hence $1 - B(u) \ge \frac{1}{5}$. As any two unit vectors are at distance at most 2, $\mathrm{dist}(u, \mathcal{C})^2 \le 4$, so $\frac{1}{20}\,\mathrm{dist}(u, \mathcal{C})^2 \le \frac{1}{5} \le 1 - B(u)$. $\square$

2.3 Edge gadgets

Let $C = \binom{k}{2}$. Encode each vertex $v \in V_i$ by an index $r_v \in [N]$; its two *signed copies* are the directions s_{r_v} and $s_{r_v+N} = -s_{r_v}$. Set

$$\tau = \frac{\rho\,(2-\mu/2)}{\beta}, \qquad \lambda = \frac{2}{\rho\mu} \qquad \text{(both positive rationals).}$$

For every edge vw of G with $v \in V_i$, $w \in V_j$, $i < j$, and for each of the four signed-copy pairs (r,q) with $r \in \{r_v, r_v+N\}$, $q \in \{r_w, r_w+N\}$, introduce the generator

$$g_{i,j,r,q} = \lambda\big(-\tau e_0 + s_r^{(i)} + s_q^{(j)}\big), \tag{6}$$

where $s_r^{(i)}$ places the 2-vector s_r in block i and 0 elsewhere. Let $E(u) = \sum [\langle g_{i,j,r,q}, u\rangle]_+$ be the sum over all edge generators.

Lemma 6. *For every candidate $c = c(t_1,\dots,t_k) \in \mathcal{C}$ and every generator (6),*

$$[\langle g_{i,j,r,q}, c\rangle]_+ = \begin{cases} 1, & \text{if } (t_i,t_j) = (r,q), \\ 0, & \text{otherwise.} \end{cases}$$

Consequently $E(c)$ equals the number of edges of G among the selected vertices; in particular $E(c) = C$ if and only if those vertices form a multicolored k-clique.

Proof. Since $\langle e_0, c\rangle = \beta$ and $\langle s_r^{(i)}, c\rangle = \rho\langle s_r, s_{t_i}\rangle$,

$$\langle g_{i,j,r,q}, c\rangle = \lambda\big(\rho\langle s_r, s_{t_i}\rangle + \rho\langle s_q, s_{t_j}\rangle - \tau\beta\big).$$

If $(t_i,t_j) = (r,q)$ then $\langle s_r, s_{t_i}\rangle = \langle s_q, s_{t_j}\rangle = 1$ and, using $\tau\beta = \rho(2-\mu/2)$,

$$\langle g_{i,j,r,q}, c\rangle = \lambda(2\rho - \tau\beta) = \lambda \cdot \tfrac{\rho\mu}{2} = \tfrac{2}{\rho\mu} \cdot \tfrac{\rho\mu}{2} = 1.$$

If instead $(t_i,t_j) \neq (r,q)$, then at least one inner product drops by at least μ by Lemma 3, while the other is at most 1; hence

$$\langle g_{i,j,r,q}, c\rangle \le \lambda\big(\rho(1-\mu) + \rho - \tau\beta\big) = \lambda\big(2\rho - \tau\beta - \rho\mu\big) = \lambda\big(\tfrac{\rho\mu}{2} - \rho\mu\big) = -1,$$

so its ReLU value is 0. Thus, for a fixed color pair (i,j), exactly the generator matching (t_i,t_j) can contribute, and it contributes 1 precisely when the selected vertices in classes i,j are the endpoints of an edge (regardless of their chosen signs, since all four signed pairs are present). Summing over the $\binom{k}{2}$ color pairs proves the claim. $\square$

Set

$$\widehat{H} = 2k^2N^2\lambda(\tau+2).$$

Lemma 7. $0 \le E(u) \le \widehat{H}$ *and* $|E(u) - E(v)| \le \widehat{H}\,\|u-v\|_2$ *for all unit* u, v.

Proof. Each edge generator has norm $\|g_{i,j,r,q}\|_2 = \lambda\sqrt{\tau^2+2} \le \lambda(\tau+2)$ (as $(\tau+2)^2 - (\tau^2+2) = 4\tau + 2 \ge 0$). For unit u, $[\langle g,u\rangle]_+ \le \|g\|_2$, so $E(u) \le \sum \|g\|_2 \le \widehat{H}$ since the total number of pair (i,j,r,q) is at most $2k^2N^2$. Since $t \mapsto [t]_+$ is 1-Lipschitz, $|\,[\langle g,u\rangle]_+ - [\langle g,v\rangle]_+\,| \le |\langle g, u-v\rangle| \le \|g\|_2\|u-v\|_2$; summing gives the Lipschitz bound with constant $\sum\|g\|_2 \le \widehat{H}$. $\square$

2.4 Objective and threshold

Define

$$\eta = \frac{1}{4\widehat{H}}, \qquad M = \left\lceil \frac{\widehat{H} + C + 1}{\frac{1}{20}\eta^2} \right\rceil + 1,$$

and we set the objective and threshold as

$$F(u) = M \cdot B(u) + E(u), \qquad T = M + C - \frac{1}{2}.$$

By construction F is the support function of a rational generator-given zonotope $Z \subseteq \mathbb{R}^{2k+1}$, as both the base function B and the edge function are support functions of zonotope. By (1), $\max_{z \in Z} \|z\|_2 = \max_{\|u\|_2 = 1} F(u)$.

Lemma 8 (Completeness)**.** *If G has a multicolored clique, then* $\max_{\|u\|_2=1} F(u) > T$.

Proof. Let the clique select vertices with signed indices $t_1, \ldots, t_k$ and let $c = c(t_1, \ldots, t_k)$. By Lemma 4, $B(c) = 1$; by Lemma 6, $E(c) = C$. Hence $F(c) = M + C > M + C - \frac{1}{2} = T$. □

Lemma 9 (Soundness)**.** *If G has no multicolored clique, then $F(u) < T$ for every unit u.*

Proof. Fix a unit vector u and consider two cases.

Case $\mathrm{dist}(u, \mathcal{C}) > \eta$. By Lemma 5, $B(u) \le 1 - \frac{1}{20}\eta^2$, so by Lemma 7,

$$F(u) \le M\left(1 - \tfrac{1}{20}\eta^2\right) + \widehat{H} = M + \widehat{H} - M \cdot \tfrac{1}{20}\eta^2.$$

Because $M > \dfrac{\widehat{H} + C + 1}{\frac{1}{20}\eta^2}$, we have $M \cdot \frac{1}{20}\eta^2 > \widehat{H} + C + 1 > \widehat{H} - C + \frac{1}{2}$. Therefore $F(u) < M + \widehat{H} - (\widehat{H} - C + \frac{1}{2}) = M + C - \frac{1}{2} = T$.

Case $\mathrm{dist}(u, \mathcal{C}) \le \eta$. Let $c \in \mathcal{C}$ be a nearest candidate, so $\|u - c\|_2 \le \eta$. Since G has no multicolored clique, $E(c) \le C - 1$ by Lemma 6. By Lemma 7, $E(u) \le E(c) + \widehat{H}\eta \le (C-1) + \widehat{H} \cdot \frac{1}{4\widehat{H}} = C - \frac{3}{4}$. Using $B(u) \le 1$ (Lemma 4),

$$F(u) = M\, B(u) + E(u) \le M + C - \tfrac{3}{4} < M + C - \tfrac{1}{2} = T. \qquad \square$$

Proof of Theorem 1. By Lemmas 8 and 9 together with (1), we have already proved $\max_{z \in Z} \|z\|_2 \ge T$ if and only if G has a multicolored k-clique. It remains to verify that this is a valid parameterized-reduction output.

Dimension and parameter. $Z \subseteq \mathbb{R}^{2k+1}$, so the parameter $d = 2k + 1$ depends only on k; the number N of vertices per class influences only the number of generators, not d.

Number of generators. The base gadget contributes 1 generator for $[0, e_0]$ and $O(N)$ generators for each of the k copies of the polygon P (which has $2N$ vertices, hence N edge-vector directions and $2N$ generators $\pm\frac{1}{2}z_\ell$), i.e. $O(kN)$; the edge gadget contributes 4 generators per edge of G, i.e. $O(k^2N^2)$. The total is polynomial in the input size.

Bit-length. The circle points s_r are rationals with numerator and denominator $O((N+1)^2)$; $\beta, \rho, \mu, \tau, \lambda$ are rationals of polynomial bit-length. Then $\lambda(\tau + 2) = O\big((4k+1)(N+1)^2\big)$ and, with $O(k^2N^2)$ edge gadgets, $\widehat{H} = \mathrm{poly}(k, N)$; hence $\eta = 1/(4\widehat{H})$ and $M = O(\widehat{H}/\eta^2) = O(\widehat{H}^3) = \mathrm{poly}(k, N)$ have $O(\log(kN))$ bit-length. All generator coordinates and the threshold $T = M + C - \frac{1}{2}$ are rationals of polynomial bit-length, computable in polynomial time. Since the comparison $\max_{z \in Z} \|z\|_2 \ge T$ can be tested against T^2 over the rationals, exactness is preserved.

Therefore, this is indeed a parameterized reduction from Multicolored Clique (parameter k, W[1]-hard by Theorem 2) to L_2-maximization over generator-given zonotopes (parameter $d = 2k + 1$). Hence the latter is W[1]-hard, and is not FPT unless FPT = W[1]. □

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